

AKASH SURIYA NARAYANAN

Chemical & Process Engineer — Process Simulation & Design

CONTACT

ak9962851601@gmail.com

+39 351 418 4915

Padova,PD, Italy

https://akashsn.com

LinkedIn

PROFILE

Chemical engineer with 2+ years of industrial experience completing an M.Sc. at the University of Padova with an industrial Aspen Plus thesis at Columbia Holding. Skilled in full-flowsheet simulation, distillation design, and thermodynamic modelling. Erasmus+ alumnus across five European universities.

LANGUAGES

English ●●●●●

Tamil ●●●●●

Hindi ●●●

Italian ●●

CORE SKILLS

- Aspen Plus (expert)
- Full-flowsheet simulation
- Distillation sizing & rating
- Thermodynamic modelling (PR, RK EOS)
- Sensitivity analysis

SOFTWARE

Aspen Plus, Power BI, ANSYS Fluent, AutoCAD, Pro/II, MATLAB, Minitab, Excel VBA

AVAILABILITY

Based in Padova, Italy

M.Sc. graduating Dec 2026

Open to roles across Italy & the EU

PROFESSIONAL EXPERIENCE

Aug 2022 - Aug 2024

Executive- Production | *Carborundum Universal Limited (CUMI), India*

‣ Led CAPEX project lifecycle (feasibility → commissioning → handoff); prepared technical business cases securing multimillion-rupee approvals from senior leadership.

‣ Led CAPEX lifecycle — feasibility, multimillion-rupee approval, vendor procurement, commissioning, operator handoff; all projects on time and within budget.

‣ Drove 75% output increase via VSM-based OEE analysis and kaizen interventions; built Power BI dashboards consolidating ERP data across 3 sites. Lean Six Sigma Yellow Belt.

EDUCATION

Sep 2024 - Mar 2027 (Expected)

M.Sc. Chemical and Process Engineering | *University of Padova, Italy*

▪ Aspen Plus simulation of distillation columns — sizing, rating, and thermodynamic modelling — in collaboration with Columbia Holding.

▪ Coursework: Multiphase Thermodynamics, Separation Unit Operations, Process Design, Process Fluid Dynamics & Simulation, Risk and Safety Analysis, Machine Learning for Process Engineers..

Jul 2018 - Sep 2022

B.Tech Chemical Engineering | *SASTRA University, India*

INTERNATIONAL PROGRAMS

Mar 2026 - May 2026

Erasmus+ Blended Intensive Program | *Hochschule Kaiserslautern, Germany*

• Decarbonisation Strategies in Process Engineering — 5-university collaboration; MVR distillation simulation in Aspen Plus; visited BASF Ludwigshafen Verbund site; Sulzer technical lecture.

KEY PROJECTS

- IBPE Plant Design (Aspen Plus, 2025–26): hierarchical design of 5,000 t/yr plant — CSTR, three-column vacuum distillation at 99.9% molar purity; full cash flow & profitability analysis.
- Toluene-to-Benzene Flowsheet (Aspen Plus, 2025–26): RStoic reactor (600°C, 30 bar), PR-EOS, RadFrac; sensitivity analysis — optimal reflux ratio 1.72 for 99.7% benzene purity; FUG vs rigorous column comparison.
- RK EOS Thermodynamic Modelling (MATLAB, 2024–25): Newton algorithm binodal reconstruction; Helmholtz, Gibbs, entropy, fugacity; validated vs CH_4, CO_2, and ethanol experimental data.
- MVR Column Simulation (Erasmus+, 2026): designed and simulated mechanical vapour recompression distillation for ethanol purification in Aspen Plus; presented at Sulzer technical lecture.

AWARDS AND RECOGNITION

▪ Star of the Quarter – FY24, Carborundum Universal Limited

▪ First Prize, Industrivia (Siphon 7.0, SVNIT Surat) & Shipwreck (STEER, SSN Chennai)

▪ Chairperson, IIChE Students' Chapter, SASTRA University (Aug 2021 – Aug 2022)

▪ Presented CFD simulation paper at SCHEMCON 2021, Indian Institute of Chemical Engineers

I authorize the processing of my personal data in accordance with the GDPR (EU Regulation 2016/679).